

Lab - 11 :Modelling stock price variations as a Bachelier-Wiener process

Krutarth Kadia (202201475)^{*} and Het Shah (202201515)[†]
*Dhirubhai Ambani Institute of Information & Communication Technology,
Gandhinagar, Gujarat 382007, India
CS302, Modelling and Simulation*

I. MODEL

The forward relative change of a stock price, S , in a finite time interval, Δt , is given by

$$\frac{\Delta S}{S} = a\Delta t + b\Delta W \quad (1)$$

Under an idealized volatility-free condition, we set $b = 0$, and then integrate it in continuous time to get a steady compounded growth of S . The integral solution of S is exponential in time,

$$S = S_0 \exp(at) \quad (2)$$

$$\Delta(\ln S) = a\Delta t \quad (3)$$

The Gaussian function, with unity added to it, as

$$f(\delta) = 1 + f_0 \exp\left[-\frac{(\delta - \mu)^2}{2\sigma^2}\right] \quad (4)$$

II. RESULTS

A.

Fig. 1 shows stock price vs time.

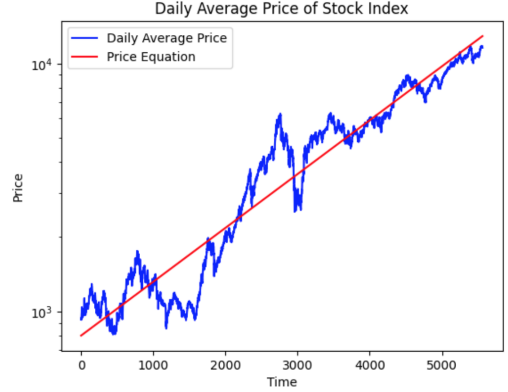


FIG. 1: The daily mean growth of the average price of the stock index, NIFTY (NSE, India). The straight line in this linear-log plot is fitted by the least-squares method, and indicates that the mean growth of S is exponential. With $b = 0$, the mean relative growth rate of stock values is a . For this plot $a = 0.05\%$ per day.

^{*}Electronic address: 202201475@daiict.ac.in

[†]Electronic address: 202201515@daiict.ac.in

B.

Fig. 2 shows daily percentage fluctuation of stock values vs time.

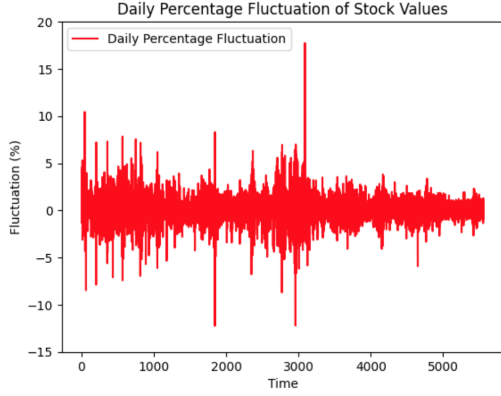


FIG. 2: The time series of the daily percentage fluctuation of prices in the stock index, NIFTY (NSE, India). The daily percentage fluctuation of prices is quantified by δ , which, over two decades, has an equal distribution of positive and negative values about $\delta = 0$.

C.

Fig. 3 shows The unnormalized frequency distribution of the daily percentage fluctuation of prices in the stock.

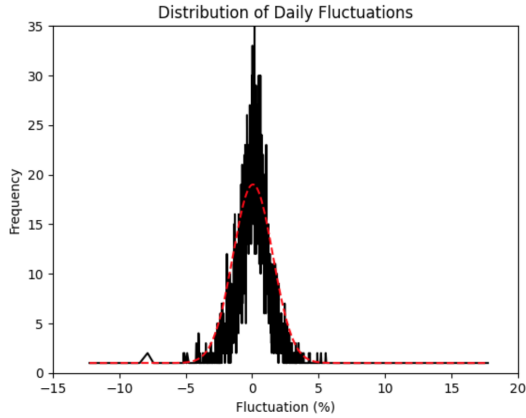


FIG. 3: The unnormalized frequency distribution of the daily percentage fluctuation of prices in the stock index, NIFTY (NSE, India). The distribution appears Gaussian, and is centred around a mean value, $\mu = 0.057$, with a standard deviation, $\sigma = 1.495$.

D.

Fig. 4 shows The growth of the monthly average of $\ln S$ vs time.

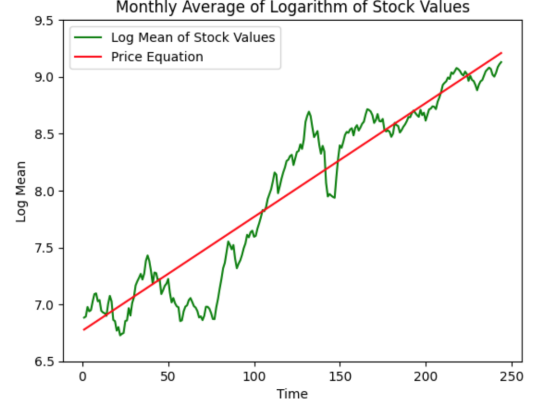


FIG. 4: The growth of the monthly average of $\ln S$ for NIFTY (NSE, India), as opposed to its daily growth in Fig(1). The straight line, showing the mean growth, is fitted by the least-squares method, and its slope is $m = 0.01$ per month.

E.

Fig. 5 shows the variance of the monthly prices, vs time.

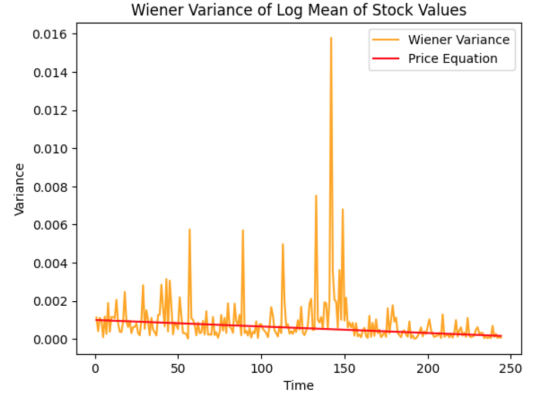


FIG. 5: The Wiener variance about the monthly average of $\ln S$ for NIFTY (NSE, India) decreases with time, (in months). The straight line, fitted by the least-squares method, traces the mean decline, with a slope of $w = 3.41 \times 10^{-6}$ per month. With $w < 0$, volatility also reduces with time.

F.

Fig. 6 shows the growth of the daily trade volume vs time.

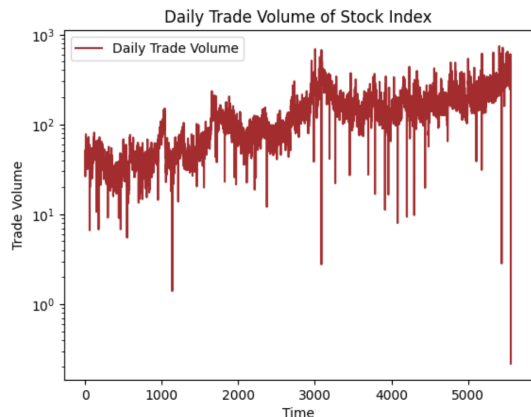


FIG. 6: The growth of the daily trade volume of the NIFTY (NSE, India) index. The straight line in this linear-log plot, fitted by the least-squares method, implies an exponential mean growth of N . The slope of the straight line, $v = 0.04\%$ per day, gives the mean relative growth rate of the daily volume of trade.

III. CONCLUSIONS

- The NIFTY market exhibits consistent growth, reflected in both stock prices and trading volumes, as indicated by the upward slope of the fitted lines in each plot.
- Over an extended period, stock values in the market tend to experience sustained growth, suggesting that prices generally increase over time.
- Simulation models offer investors and traders valuable insights into market behavior, enabling them to make informed decisions and manage risks more effectively.

[1] Abhin Kakkad, Harsh Vasoya and Arnab K. Ray, '*Regularities in stock markets*'.